

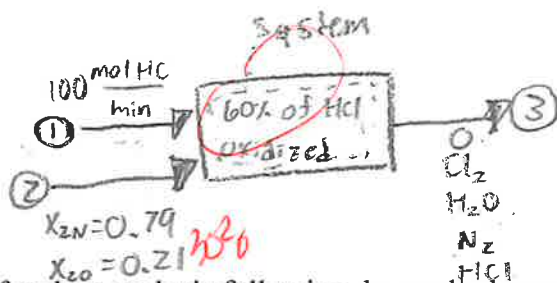
To receive full credit, you must show all your work.

- (1) (55 pts.) Hydrogen chloride (HCl) and oxygen (O₂) are reacted in the gas phase to form chlorine (Cl₂) and water (H₂O). A gas feed stream entering the reactor contains 100 mol/min of HCl and air (79 mol% N₂, 21 mol% O₂). The air in the stream contains 30% more O₂ than stoichiometrically required to oxidize all the HCl. In the reactor, 60% of the HCl is oxidized. The product stream contains all the Cl₂ and H₂O produced, the inert N₂, and the unreacted portion of the HCl and O₂.

- (a) Write a balanced chemical reaction equation for this reaction. Use any method you like to balance the equation. (5 pts.)



- (b) Draw and label a block flow diagram for the process. (5 pts.)



- (c) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Give values for 4a, 4b, 5a, 5b, 5c, and 5d. Compute the DOF. (5 pts.)

$$\begin{array}{l} 4a: 8 \\ 4b: 1 \\ \hline 9 \end{array}$$

$$\begin{array}{l} 5a: 1 \\ 5b: (1+1)=2 \\ 5c: 1 \\ 5d: 5 \\ \hline 9 \end{array}$$

$$\begin{aligned} \text{DoF} &= (4a + 4b) - (5a + 5b + 5c + 5d) \\ &= 9 - 9 \\ &= 0, \text{ the problem is well} \\ &\quad \text{Posed} \end{aligned}$$

- (d) Calculate the molar flow rates of the O₂ and N₂ in the feed stream to the reactor. (10 pts.)

30% more O₂ than stoich. required

$$\dot{n}_{\text{O}_2} = 100 \frac{\text{mol HCl}}{\text{min}} \times (1.3) \times \frac{1 \text{ mol O}_2}{4 \text{ mol HCl}} = 32.5 \frac{\text{mol O}_2}{\text{min}}$$

$$\frac{\dot{n}_2 \cdot x_{\text{O}_2}}{x_{\text{O}_2}} = \frac{32.5 \frac{\text{mol O}_2}{\text{min}}}{x_{\text{O}_2}} \cdot \left\{ x_{\text{O}_2} = 0.21 \right\} = 154.76 \frac{\text{mol tot}}{\text{min}} = \dot{n}_2$$

$$\dot{n}_2 \cdot x_{\text{N}_2} = \dot{n}_{\text{N}_2} = 154.76 \frac{\text{mol tot}}{\text{min}} \times 0.79 = 122.26 \frac{\text{mol N}_2}{\text{min}}$$

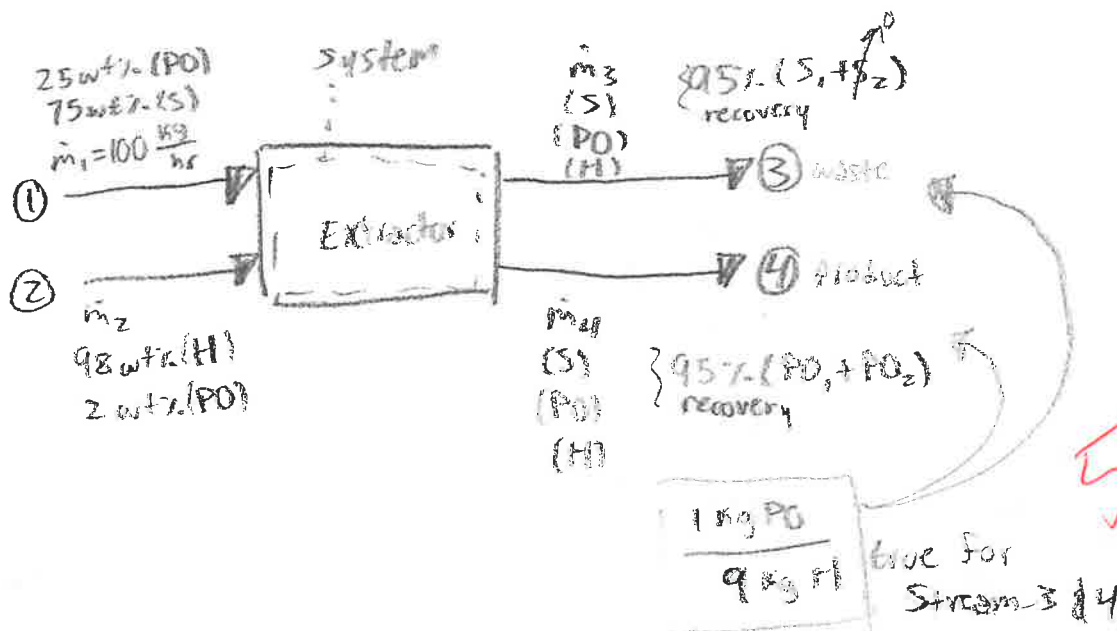
Feed Stream molar flow rate:

Acid (HCl) (mol/min)	Oxygen (O ₂) (mol/min)	Nitrogen (N ₂) (mol/min)
100	32.5	122.26

Given

(2) (45 pts.) Hexane can be used to extract peanut oil from ground peanuts. There are two streams entering the extractor unit: (1) the feed stream $\dot{m}_1 = 100 \text{ kg/hr}$ containing ground peanuts which has a composition of 75 wt% solids (S) and 25 wt% peanut oil (PO), and (2) a stream $\dot{m}_2$ of 98 wt% hexane (H) and 2 wt% peanut oil. There are two streams exiting the extractor unit: (3) a waste stream $\dot{m}_3$ containing solids, peanut oil and hexane, and (4) the product stream $\dot{m}_4$ which also contains solids, peanut oil and hexane. The waste stream recovers 95% of the solids entering the extractor while the product stream recovers 95% of the peanut oil entering the extractor. The equilibrium solubility of peanut oil in hexane is 1 kg of peanut oil to 9 kg of hexane. Assume that this equilibrium has been achieved in both exit streams.

(a) Draw a block flow diagram of the process and label all the streams. (5 pts.)



(b) Solve the appropriate mass balances along with the recovery equation above to calculate all the species and total mass flow rates in kg/hr. Fill in the table below with your answers. (40 pts.)

$$\textcircled{1} \quad \dot{m}_{1PO} = 0.25 \dot{m}_1, \quad \left\{ \dot{m}_1 = 100 \frac{\text{kg}}{\text{hr}} \right\} = 25 \frac{\text{kg PO}}{\text{hr}}$$

$$\dot{m}_{1S} = 0.75 \dot{m}_1 = 75 \frac{\text{kg S}}{\text{hr}}$$

$$\text{S balance: } \dot{m}_{1S} = \dot{m}_{3S} + \dot{m}_{4S} \Rightarrow 75 \frac{\text{kg S}}{\text{hr}} = \dot{m}_{3S} \cdot 0.95 + \dot{m}_{4S} \Rightarrow \dot{m}_{4S} = 75 - 71.25$$

$$\dot{m}_{3S} = \dot{m}_{1S} \cdot 0.95 = 71.25$$

$$\dot{m}_{4S} = 3.75 \frac{\text{kg S}}{\text{hr}}$$

$$\text{mass balance: } \dot{m}_1 + \dot{m}_2 = \dot{m}_3 + \dot{m}_4$$

$$\text{PO balance: } \dot{m}_{1PO} + \dot{m}_{2PO} = \dot{m}_{3PO} + \dot{m}_{4PO}$$

$$0.25 \dot{m}_1 + 0.02 \dot{m}_2 = \dot{m}_{3PO} + 0.95(0.25 \dot{m}_1 + 0.02 \dot{m}_2)$$

$$\text{H balance: } \dot{m}_2 \cdot 0.98 = \dot{m}_{3H} + \dot{m}_{4H}$$

$$\frac{\dot{m}_{3PO}}{\dot{m}_{3H}} = \frac{1}{9}, \quad \frac{\dot{m}_{4PO}}{\dot{m}_{4H}} = \frac{1}{9}$$

(Problem 1 Continued)

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(e) Calculate the species molar flow rates for all five components in the product stream in mol/min. Show all your calculations and write your final answers in the space provided below. (30 pts.)

Species	$\frac{\text{mol}}{\text{min}}$ $\dot{n}_1$	x_1	$\frac{\text{mol}}{\text{min}}$ $\dot{n}_2$	x_2	$\frac{\text{mol}}{\text{min}}$ $\dot{n}_3$	x_3
Cl_2	100	1	—	—	30	
H_2O	—	—	—	—	30	
N_2	—	—	122.26	0.79	122.26	
HCl	—	—	—	—	40	
O_2	—	—	32.5	0.21	17.5	
tot	100	1	154.76	1	239.76	1

$$\begin{array}{r} 1 \mid 54 \\ 2 \mid 31 \\ \hline 1 \mid 85 \end{array}$$

60% HCl oxidized $\rightarrow 60 \frac{\text{mol HCl}}{\text{min}} \text{ used} \times \frac{1 \text{ mol } \text{O}_2}{4 \text{ mol HCl}} \rightarrow 15 \frac{\text{mol } \text{O}_2}{\text{min}} \text{ used} = \dot{n}_{\text{O}_2, \text{used}}$ Cons.

$$\dot{n}_{\text{O}_2} = \dot{n}_{\text{O}_2} - \dot{n}_{\text{O}_2, \text{used}} = 32.5 - 15 = 17.5 \frac{\text{mol } \text{O}_2}{\text{min}}$$

$$60 \frac{\text{mol HCl}}{\text{min}} \text{ used} \times \frac{2 \text{ mol } \text{Cl}_2}{4 \text{ mol HCl}} = 30 \frac{\text{mol } \text{Cl}_2}{\text{min}} = \text{Cl}_2 \text{ generated}$$

$$\text{or } 60 \frac{\text{mol HCl}}{\text{min}} \text{ used} \times \frac{2 \text{ mol } \text{H}_2\text{O}}{4 \text{ mol HCl}} = 30 \frac{\text{mol } \text{H}_2\text{O}}{\text{min}} \text{ generated}$$

No Cl_2 or H_2O to begin with, so generation term is the final product term

$$\dot{n}_2 x_{2N} = \dot{n}_3 x_{3N} = \dot{n}_{3N}, \{ \dot{n}_2 = 154.76, x_{2N} = 0.79 \} = 122.26 \frac{\text{mol } \text{N}_2}{\text{min}}$$

$$\dot{n}_{\text{HCl}} = \dot{n}_{\text{HCl}} - \dot{n}_{\text{HCl, consumed}} = \dot{n}_{\text{HCl}} - \dot{n}_{\text{HCl}} \cdot 0.6 = 40 \frac{\text{mol HCl}}{\text{min}}$$

30

Product Stream molar flow rate:

Acid (HCl) (mol/min)	Oxygen (O_2) (mol/min)	Nitrogen (N_2) (mol/min)	Chlorine (Cl_2) (mol/min)	Water (H_2O) (mol/min)
40	17.5	122.26	30	30

54

(Problem 2 Continued)

$$\frac{0.05(\dot{m}_{1PO} + \dot{m}_{2PO})}{\dot{m}_{3H}} = \frac{1}{9} \Rightarrow \dot{m}_{3H} = 0.45(\dot{m}_{1PO} + \dot{m}_{2PO})$$

$$\frac{0.95(\dot{m}_{1PO} + \dot{m}_{2PO})}{\dot{m}_{4H}} = \frac{1}{9} \Rightarrow \dot{m}_{4H} = 8.55(\dot{m}_{1PO} + \dot{m}_{2PO})$$

from H balance: $0.98 \dot{m}_2 = (0.45 + 8.55)(\dot{m}_{1PO} + \dot{m}_{2PO}) \Rightarrow 0.98 \dot{m}_2 = 9 \dot{m}_{1PO} + 9 \dot{m}_{2PO}$

from PO balance:

$$\frac{\dot{m}_{1PO} + \dot{m}_{2PO}}{0.95(\dot{m}_{1PO} + \dot{m}_{2PO})} \times \dot{m}_{3PO} = 1.053 \frac{\text{kg PO}}{\text{hr}} \quad \text{?}$$

$$\dot{m}_{4PO} = \dot{m}_{1PO} + \dot{m}_{2PO} - \dot{m}_{3PO} = 29.572 \frac{\text{kg PO}}{\text{hr}}$$

$$\frac{\dot{m}_{3PO}}{\dot{m}_{3H}} = \frac{1}{9} \Rightarrow 9 \dot{m}_{3PO} = \dot{m}_{3H} \Rightarrow \dot{m}_{3H} = 9.477 \frac{\text{kg}}{\text{hr}}$$

$$9 \dot{m}_{4PO} = \dot{m}_{4H} = 266.148 \frac{\text{kg H}}{\text{hr}}$$

$$\dot{m}_3 = \dot{m}_{3PO} + \dot{m}_{3S} + \dot{m}_{3H} = 81.78 \frac{\text{kg}}{\text{hr}}$$

$$\dot{m}_4 = \dot{m}_{4PO} + \dot{m}_{4S} + \dot{m}_{4H} = 299.47 \frac{\text{kg}}{\text{hr}}$$

$$\Rightarrow 0.98 \dot{m}_2 = 9 \dot{m}_{1PO} + 9(\dot{m}_2 - 0.02)$$

$$\Rightarrow 0.8 \dot{m}_2 = 9 \dot{m}_{1PO}$$

$$\Rightarrow \dot{m}_2 = \frac{9}{0.8} \dot{m}_{1PO}$$

$$\Rightarrow \dot{m}_2 = 281.25 \frac{\text{kg}}{\text{hr}}$$

$$\dot{m}_{2PO} = 0.02 \cdot 281.25 \frac{\text{kg}}{\text{hr}} = 5.625 \frac{\text{kg}}{\text{hr}}$$

$$\dot{m}_2 - \dot{m}_{2PO} = \dot{m}_{2H}$$

$$\dot{m}_{2H} = 275.625 \frac{\text{kg}}{\text{hr}}$$

6
10
5
5

31

Species \ Stream#	1	2	3	4
PO (kg PO/hr)	25 ✓	5.625 ✓	1.053 ✗	29.572 ✗
S (kg S/hr)	75 ✓	0	71.25 ✓	3.75 ✓
H (kg H/hr)	0 ✓	275.625 ✓	9.477 ✗	266.148 ✗
Total (kg/hr)	100 ✓	281.25 ✓	81.78 ✓	299.47 ✓

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3 + \dot{m}_4 \quad \odot$$